

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Architected and shipped production **GenAI** infrastructure in **Python/C++** spanning **OpenAI, Anthropic**, and **Google** models, orchestrating **multimodal** generation, tool-use, and multi-step agent workflows at **99%+ availability**.
- Designed and owned an agentic **evaluation** and release-gating platform, automatically testing model quality, tool-use correctness, and regression behavior, blocking **15%** of failing releases before production.
- Built and optimized real-time **LLM/ML** serving infrastructure on **AWS SageMaker**, achieving **sub-200ms p99 latency** by profiling inference paths and resolving model and systems bottlenecks.
- Drove architecture tradeoffs across frontier models and **GenAI** techniques, evaluating quality, latency, reliability, cost, **multimodal** capability, and tool-use behavior to select production approaches per application domain.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Led the redesign of production **AI retrieval infrastructure**, improving recall **61% to 89%** on a 2,000-query evaluation set through **hybrid dense-sparse retrieval**, re-ranking, and systematic error analysis.
- Built and evaluated **multimodal** and **generative-model** pipelines in **PyTorch**, spanning **CNN/ResNet, UNet** segmentation, and **Vision Transformer**-based architectures, with benchmarking across model quality, robustness, and inference behavior.
- Owned model-quality and **evaluation infrastructure** for production AI systems, tracing regressions across data, preprocessing, retrieval, and model-execution layers, driving fixes from offline evaluation through production validation.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Architected and shipped **25+ C++/Java** distributed services serving **1M+ users** at **99.9% availability**, owning system-design decisions spanning service boundaries, APIs, and reliability.
- Reduced production escalations **60%** by defining standardized API contracts, timeout/retry semantics, and failure-handling patterns across 25+ production services.
- Designed fault-tolerant distributed-service architectures with explicit timeout budgets, idempotent operations, bounded retries, and graceful degradation to prevent downstream failure propagation.
- Owned production debugging across application, network, and service layers, using logs, metrics, and root-cause analysis to resolve recurring reliability issues, feeding fixes back into platform standards via peer design and code reviews.

PROJECTS

HoG-ViT (M.S. Thesis)

github.com/ganeshhgupta/HoG-ViT

- M.S. thesis project: designed a **Vision Transformer** variant fusing **Histogram of Oriented Gradients** edge/gradient features directly into patch embeddings, giving the model an explicit geometric prior before self-attention; implemented the full training and evaluation pipeline in **PyTorch**, benchmarking against a standard **ViT** baseline and demonstrating improved robustness to lighting and pose variation on image classification.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- LangGraph** multi-agent system automating production incident diagnosis via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Production **GenAI** evaluation platform with rubric-based scoring, hallucination detection, and regression alerting across model versions; **FastAPI** backend, **PostgreSQL**.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, C++, Java, TypeScript, SQL

Fundamentals: Data Structures and Algorithms, Operating Systems, Database Systems, Computer Networks

GenAI/ML: LLMs, Multimodal Pipelines, Prompt Engineering, RAG, ML Infrastructure, Model Evaluation/Optimization, PyTorch, TensorFlow

Computer Vision: CNN, ResNet, UNet, Vision Transformers, Image Segmentation

Web & Infra: HTML5, CSS, React, AWS (SageMaker, Lambda, EC2), GCP, Docker, Kubernetes, CI/CD

Certifications: AWS Certified Developer Associate, Google Cloud Professional Machine Learning Engineer